

User Guide

Engineering Data Backbone — Material Master, BOM, Procurement, Inventory & Reports · Version 1.0

This guide explains **what the platform does and how your team uses it day to day**. It is written for everyone at Innopolis — engineers, purchase, stores, commercial and management. No technical knowledge is needed.

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1. What is the PLM Platform?

The Innopolis PLM (Product Lifecycle Management) platform is the company's single, trusted source of engineering and procurement data — the **Engineering Data Backbone**. Instead of materials, BOMs, vendor lists and purchase records living in scattered spreadsheets, everything lives in one connected system where each piece of data is entered once and reused everywhere.

The platform follows a project from first enquiry all the way to delivered goods, across six connected modules:

#	Module	What it does
1	Material Master	The catalogue of every material, with an intelligent code. The foundation for everything else.
2	Project & BOM	Bills of Materials for each project, with a formal review-and-approval workflow.
3	Vendor Database	Your approved suppliers and their details.
4	Procurement	Request for Quotation → compare quotes → Purchase Order → goods receipt.
5	Inventory	Live stock balances across warehouses, with low-stock alerts.
6	Reports & Analytics	Dashboards for spend, vendor performance, stock value and project cost.

Golden rule: data flows one way and is never re-typed. A material defined once in the Master is picked into a BOM; the approved BOM feeds an RFQ; the winning quote becomes a Purchase Order; the received goods update Inventory; and every step feeds the Reports. Enter it right once, and it stays right everywhere.

2. Getting Started & Logging In

1. **Log in** with the email and password given to you by your administrator.
2. **First-time login:** if you were issued a temporary password, the system will require you to set a new password before you can do anything else. This is a one-time security step.
3. Once logged in you'll see only the areas your role is allowed to use (see the next section).

If you're asked to change your password and can't continue — that's expected. The platform locks every other screen until a new password is set. Set it once and you're in.

Sessions expire after a period of inactivity for security; simply log in again if that happens. Forgot your password? Your administrator can reset it for you — they cannot see your old one.

3. Roles — Who Does What

Every user has one **role**. Your role decides what you can *view* and *change*. Everyone can view the shared data they need; only the responsible role can create or modify it. The **Administrator** can do everything.

Role	Responsible for	Can create / edit
Administrator	Users, configuration & all master data	Everything
Engineering	Materials (parts), Projects & BOMs	Parts, projects, BOMs and BOM lines; submits BOMs for review
Commercial	Commercial review of BOMs & pricing	Approves BOMs at the commercial review stage
Purchase	Vendors & buying	Suppliers, RFQs, Purchase Orders; releases approved BOMs to purchasing
Stores	Warehouses & stock	Warehouses, stock adjustments & transfers, goods receipt
Management	Oversight	Views dashboards, reports & analytics

The classification data that underpins the codes (categories, subtypes, specifications, grades and units) is maintained by the Administrator so the coding scheme stays consistent company-wide.

4. The Material Code Explained

Every material carries an **intelligent code** — a code that tells you what the item *is* just by reading it. The code has four parts, each drawn from a controlled list:

TT
–
SS
–
MM
–
DDDD

Segment	Means	Example
TT	Type — the top-level category (Material Category)	Fasteners
SS	Subtype — a subdivision of the type	Bolts
MM	Major Spec — the primary specification	M10
DDDD	Detail Spec — the finer detail that makes the item unique	×50mm A2

Because every segment comes from a maintained master list, codes are consistent, searchable and never duplicated. This is why the **Material Master must be set up first** — it defines the building blocks for every code.

5. Module 1 · Material Master

The Material Master is the catalogue at the heart of the platform. It is built in layers, from broad classification down to the individual material.

The building blocks

Item	What it is	Maintained by
Category (Material Type)	Top-level classification — the TT in the code (e.g. Fasteners, Raw Metal). Each carries a default unit of measure.	Administrator
Subtype	A subdivision of a category — the SS (e.g. Bolts under Fasteners).	Administrator
Major Spec	Primary specification — the MM (e.g. thread size, material grade).	Administrator
Grade	Material grade / quality reference used on parts.	Administrator
Unit	Units of measure (Nos, kg, m, ...) used across the system.	Administrator
Part (Material)	The actual catalogued material, with its intelligent code, specs, grade, unit, sourcing (Make / Buy / Standard) and lifecycle.	Engineering

Category Management (worked example)

Categories are the first thing set up because they anchor the whole coding scheme. For each category the administrator records:

- **Name** — e.g. "Fasteners" (must be unique).
- **Type code** — the short **TT** code, e.g. "FS" (must be unique; stored in capitals).
- **Default unit of measure** — the unit new materials in this category start with (e.g. "Nos").
- **Active / Inactive** — inactive categories stay in the records but are hidden from new-material forms.

Retiring vs deleting: to stop a category from being used going forward, set it to *inactive* rather than deleting it. That keeps all history intact while removing it from selection lists. Deletions are "soft" — the record is hidden, never truly erased.

Names and codes must be unique. If you try to reuse a name or type code that already exists, the system will stop you so the catalogue stays clean.

Parts — the materials themselves

A **Part** is a real, orderable material. Engineering creates parts by combining the building blocks above; the system assembles the intelligent code automatically. Each part also records its **lifecycle** stage — Concept → In Design → In Review → Released → Production → Obsolete — so everyone knows whether a material is ready to be used in production.

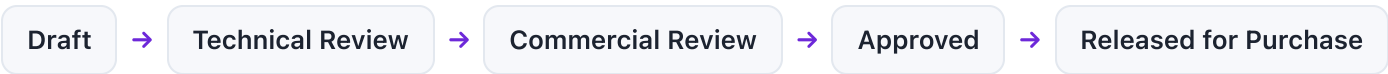
6. Module 2 · Project & BOM

A **Project** represents a customer job and moves through its own lifecycle (Enquiry → Technical Evaluation → Quotation → Project Order → Detailed Engineering → Final BOM → Purchase Release → Procurement → Fulfilment → Completed). Projects and BOMs are numbered automatically (for example **INP-2026-001**).

A **BOM (Bill of Materials)** is the structured list of every material and quantity needed to build the project's product. Engineering builds the BOM by adding **lines**, each pulling a material straight from the Master (so specs, unit and cost come across automatically). The platform continuously rolls up totals — total value, number of unique materials, count of critical items and long-lead items — as lines are added.

The BOM approval workflow

A BOM cannot go straight from an engineer's desk to purchasing. It passes through a controlled, auditable approval flow:



Stage	What happens	Who advances it
Draft	Engineering builds and edits the BOM lines. This is the <i>only</i> stage where lines can be changed.	Engineering
Technical Review	The technical content is checked.	Engineering
Commercial Review	Pricing and commercial aspects are checked.	Commercial
Approved	The BOM is signed off and ready to be released.	Purchase (releases it)
Released for Purchase	Locked and handed to Procurement. This stage is final.	—

- **Advancing** moves the BOM to the next stage. **Rejecting** a review sends it back to Draft for rework.
- A BOM with no lines cannot leave Draft, and a Released BOM can no longer be edited.

- Every transition is recorded in an **audit trail** — who moved it, when, and to which stage.

BOM analysis: at any point you can analyse a BOM's cost broken down by category, vendor, lead time or make-vs-buy — useful for planning procurement and spotting long-lead or high-cost items early.

7. Module 3 · Vendor Database

The Vendor Database is the Purchase team's list of approved suppliers. Each supplier record holds contact and commercial details plus an approval **status**:

Status	Meaning
Preferred	First-choice supplier.
Approved	Cleared for use.
Conditional	Usable with conditions / caution.
Under Review	Being evaluated — not yet cleared.

Suppliers are linked to materials and to quotations, so the platform can later report on how much you spend with each vendor and how they perform on delivery.

8. Module 4 · Procurement

Procurement turns an approved BOM into actual purchased goods, through a clear four-step flow:



Step 1 · Request for Quotation (RFQ)

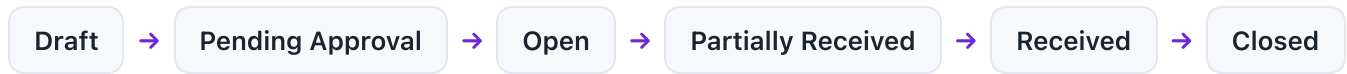
Purchase raises an **RFQ** — often generated directly from a released BOM, so the items and quantities come across automatically. RFQs can be organised different ways (vendor-wise, category-wise, package-wise, single item or bulk). The RFQ starts as *Draft*, then is *Sent* to vendors.

Step 2 · Quotations & comparison

Vendors submit **quotations** with per-line pricing (one quotation per vendor). As quotes arrive the RFQ moves to *Quotes In*. The platform then **compares** them and ranks the offers — the cheapest gets rank 1 — so the buyer can make an objective decision. The buyer **awards** the winning quotation; the winner is marked *Awarded* and the rest *Rejected* automatically.

Step 3 • Purchase Order (PO)

The awarded quotation becomes a **Purchase Order** (POs are auto-numbered). A PO moves through its own status lifecycle:



A PO can also be **Cancelled**. Line totals (quantity × unit price) are calculated for you.

Step 4 • Goods Receipt

When goods arrive, Stores (or Purchase) records a **receipt** against the PO, entering how much was received and how much was accepted vs rejected after inspection. The PO updates to *Partially Received* or *Received* accordingly.

Rejected goods never enter stock. Only accepted quantities are posted to inventory, keeping your stock figures trustworthy. When a receiving warehouse is specified, accepted goods flow straight into Inventory automatically.

9. Module 5 • Inventory

Inventory tracks how much of each material you hold, and where. Stock is held across **warehouses** (Distribution, Manufacturing, Buffer or Transit) and every change is written to a permanent **stock ledger** — an append-only history you can always audit.

WHAT STORES CAN DO

- **Opening stock** — record the starting quantity of a material.
- **Adjust** — correct a balance (e.g. after a physical count) or record wastage.
- **Transfer** — move stock from one warehouse to another.
- **Receive** — accepted goods from a PO arrive as stock automatically.

WHAT EVERYONE CAN SEE

- **Stock balances** — current quantity of every material by warehouse.
- **Movements** — the full history of ins and outs.
- **Low-stock alerts** — materials at or below their reorder point, so you can replenish in time.

Each material's availability is kept in step automatically — *In Stock*, *Low Stock* or *Out of Stock* — based on its quantity and reorder point.

10. Module 6 · Reports & Analytics

The Reports module turns everyday activity into insight. Management and other roles can view:

Report	Answers the question...
Dashboard	What's the overall health of projects, procurement and stock right now?
Purchase value	How much are we purchasing, over time?
Vendor performance	Which suppliers deliver well?
Stock value	How much money is tied up in inventory?
Vendor spend	How much do we spend with each vendor?
Project cost	What is each project costing us?

Because every module feeds the same data backbone, these reports are always based on live, consistent figures — no manual consolidation required.

11. Glossary

Term	Meaning
PLM	Product Lifecycle Management — managing product data from concept to delivery.
Material Master	The central catalogue of all materials and their classification.
Intelligent code	The TT-SS-MM-DDDD code that describes a material at a glance.
Part	An individual catalogued material.
BOM	Bill of Materials — the list of items and quantities to build a product.
Lifecycle	The stages a material or project passes through over its life.
RFQ	Request for Quotation — asking vendors to quote prices.
Quotation	A vendor's priced offer in response to an RFQ.
PO	Purchase Order — a formal order placed with a vendor.
Goods receipt	Recording items physically received against a PO.
Reorder point	The stock level at which a material should be replenished.
Soft delete	Hiding a record from view while keeping it in the database for history.
Audit trail	A permanent record of who changed what, and when.